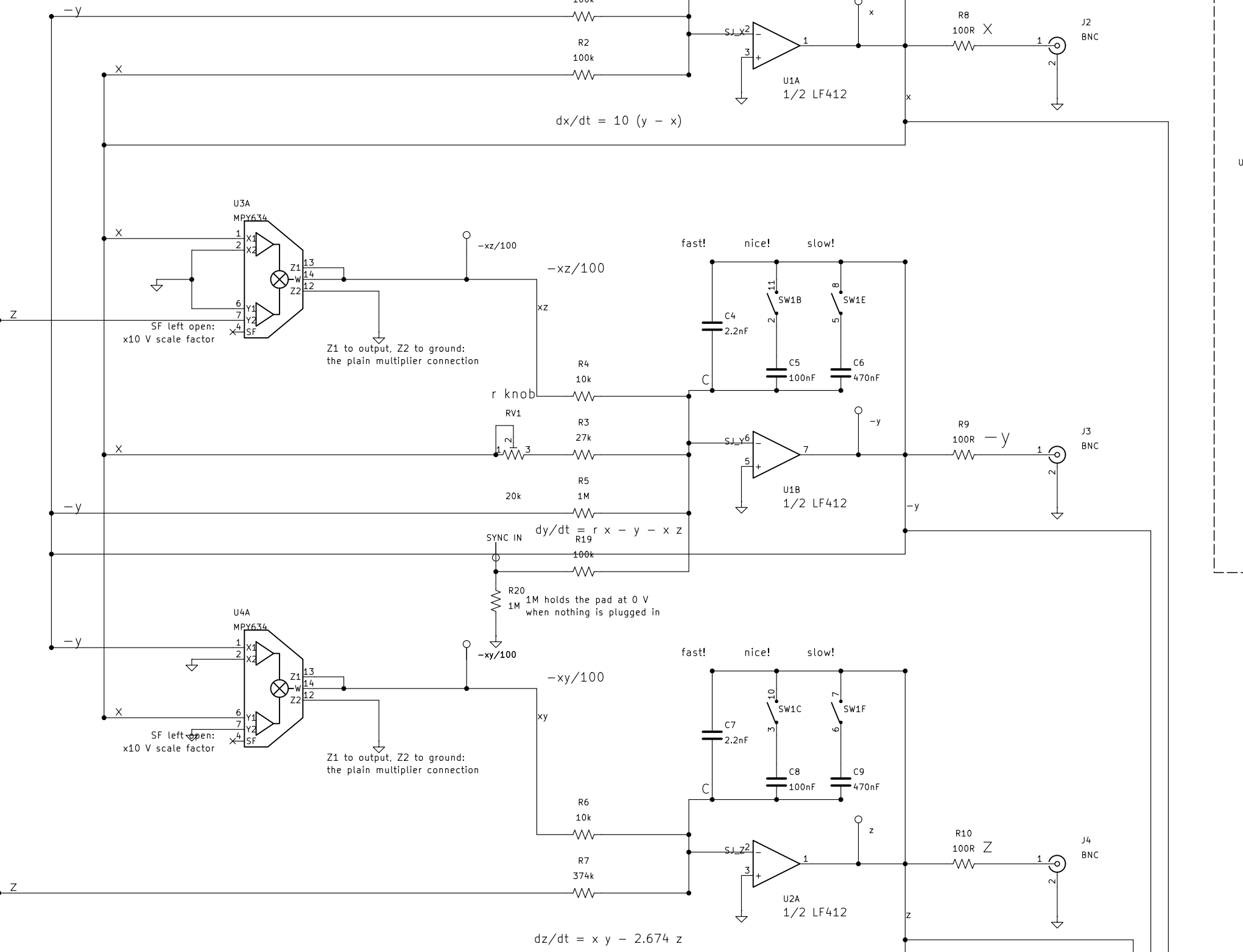


C = 0.47 uF (slow!)
= 0.1 uF (nice!)
= 2000 pF (fast!)

...built here as 2.2 nF always fitted, plus 100 nF and 470 nF switched in by SW1. See the notes at the right of the sheet.

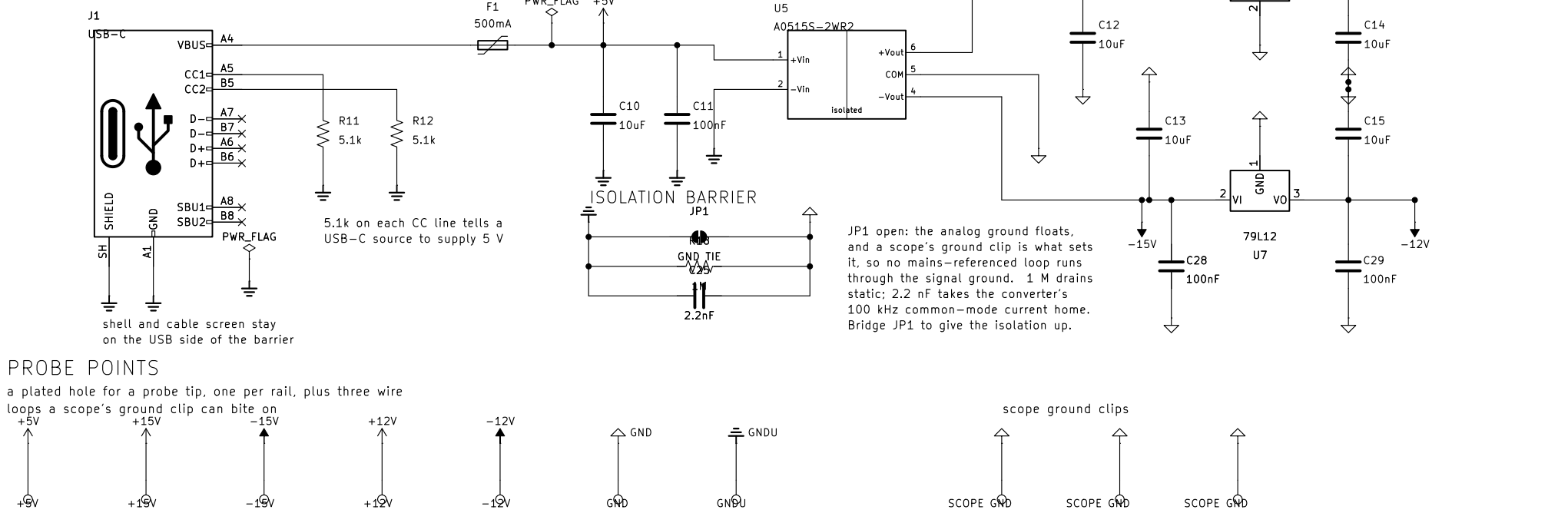


CHAOS LAMP

One lamp, three colours, driven from the three outputs.

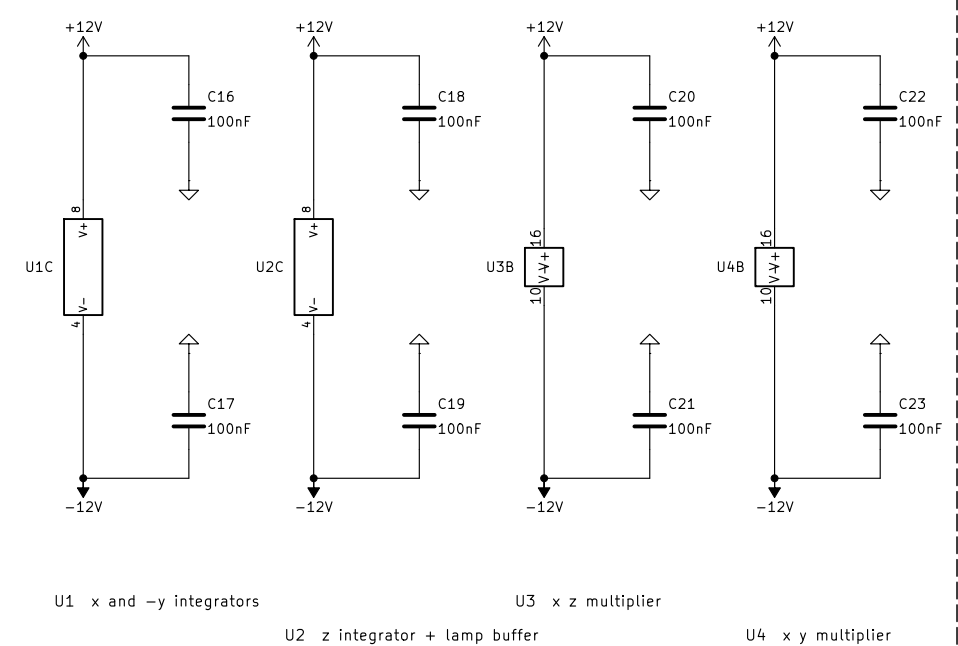
+12 V x 12k / (12k + 33k) = +3.2 V, buffered by U2B -- the half of the LF412 Paul never needed -- and held on the lamp's common anode. Each die then lights when its own signal goes low, and the three come within 5 % of the same peak brightness, so the colour wanders through mixed hues rather than blinking between saturated primaries. The taps are on the output side of each integrator, ahead of the 100 ohm series resistor, so no lamp current flows in the BNC output impedance.

POWER: USB-C 5 V in, +/-12 V out
about 25 mA per rail; the 2 W module is rated +/-66 mA

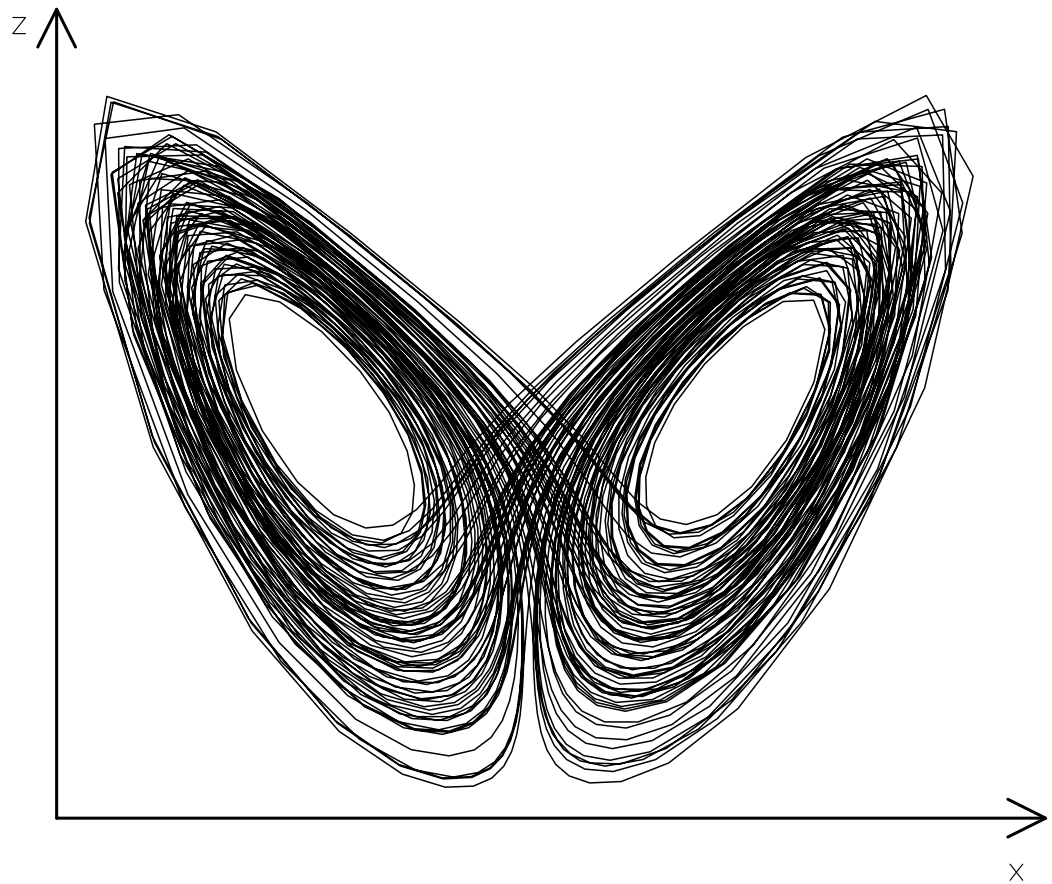


SUPPLY PINS AND BYPASSING

one 0.1 uF at every supply pin, right at the package



Hook x and z to a scope in X-Y and this is what you get: the attractor's characteristic owl's face. The trace below is the real solution of these equations, integrated at s = 10, r = 28, b = 8/3.



HOW THIS CIRCUIT SOLVES THE EQUATIONS

$$\begin{aligned} dx/dt &= s(y - x) & s &= 10 \\ dy/dt &= r x - y - x z & r &= 28 \\ dz/dt &= x y - b z & b &= 8/3 \end{aligned}$$

Each op-amp is an integrator with the terms of one derivative summed at its inverting input, so the weight of every term is simply 1 MEG / R :

R2 = R1 = 100k -> 1M/100k = 10.00 = s (and the -x damping)
R3 + RV1 = 27k + (20k..0) -> 1M/47k .. 1M/27k = 21.3 .. 37.0 = r
R5 = 1M -> 1M/1M = 1.00 = the -y term
R7 = 374k -> 1M/374k = 2.674 = b (8/3 = 2.667, 0.3 % high)
R4 = R6 = 10k -> 1M/10k = 100 cancels the multipliers' /100

Voltages carry the dimensionless variables at 0.1 V per unit, so x swings +/-2 V, y +/-2.7 V and z runs 0 to 4.8 V. An MPY634 forms A*B/10, so the product of two 0.1 V/unit signals comes out 100x too small -- which is exactly what the 10k weighting resistors R4 and R6 put back. The minus signs on both products are free: U3 and U4 simply have their differential inputs swapped.

The time scale is tau = 1 MEG x C, and C is switchable:

speed	1	2	3	4	5	6	C	tau = 1M x C
fast!	off	off	off	off	off	off	2.2 nF	2.2 ms
nice!	ON	ON	ON	off	off	off	102 nF	102 ms
slow!	off	off	off	ON	ON	ON	472 nF	472 ms
glacial!	ON	ON	ON	ON	ON	ON	572 nF	572 ms

All four are printed on the back of the board, next to the switch.

Paul's original ran on a bench +/-15 V supply and used 2000 pF, 0.1 uF and 0.47 uF. Here the three values are built up in parallel because no supplier stocks a 3-pole 3-position switch; 2.2 nF is always fitted. Both banks ON adds a fourth speed Paul never had, slow enough to watch by eye.

MOUNTING

Four M3 holes, one per board corner. No electrical function, but with three BNCs along one edge the cables lever on the board, so it wants standoffs rather than a bare bench.



THE r KNOB

RV1 is the only control that changes the shape of the attractor. Clockwise raises r, and 310 degrees of screw covers 21.3 to 37.0:

27 %	r = 24.06	below this, the trace spirals into one wing
33 %	r = 24.74	above this, both fixed points are unstable
56 %	r = 28	Lorenz's own value -- the owl's face
100 %	r = 37	wings still there, orbit tighter and faster

Between 27 % and 33 % -- eighteen degrees of screw -- chaos and the two fixed points are both stable, and which one you land in depends on where you started. Elsewhere r moves about 0.05 per degree, so a few percent of knob is a visible change and you can still find your way back to a setting you liked.

The knob is wired as a rheostat, terminal 3 and the wiper, with terminal 1 tied to the wiper. That short is the point: if grit ever lifts the wiper, the whole 20k track still bridges the branch, so r goes to its minimum instead of the r x term disappearing. Wipers fail open, and this one is not trusted not to.

SYNCHRONISING TWO BOARDS

Run a wire from one board's x output to a second board's SYNC IN pad. Within a few seconds the second board is drawing the first board's trajectory, whatever state it was in when you connected it. Pull the wire off and the two drift apart again: the same equations, states that agree to a few millivolts, and completely different futures within a minute. That is the demonstration, and it needs no switch.

R19 = 100k sets the coupling to 1M/100k = 10. It lands on the r x term, so it also adds 10 to the receiving board's r: set the driver to r = 32 and the receiver to r = 22 and both are solving the same equations. That offset is why the knob spans more than 10.

x is the channel that gets a pad because the knob can absorb the offset. -y would lock too, into the top of R1, but there is no knob for s to absorb it with. z never locks at any strength -- the x, y pair driven by z keeps a positive conditional Lyapunov exponent, the one case Pecora and Carroll's 1990 paper says must fail. Try it.